ROOTEDAI

Adhi Compliance

AI Compliance, Automated.

DETAILED ARCHITECTURE DOCUMENT

RootedAI — Engineering Intelligence & Enterprise Automation

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Table of Contents

1. Executive Summary

2. System Architecture Overview

3. Frontend Architecture

- 3.1 Application Structure & Key Pages
- 3.2 Component Hierarchy & State Management
- 3.3 Design System

4. Backend Architecture

- 4.1 FastAPI Service Layer & API Routes
- 4.2 Authentication & Authorization
- 4.3 Background Task Processing

5. Data Architecture

- 5.1 Complete Database Schema
- 5.2 Vector Store & Data Flow

6. AI/ML Architecture

- 6.1 LLM Integration — Google Gemini
- 6.2 Risk Classification Engine
- 6.3 Bias Detection & Embedding Pipeline

7. Compliance Engine Architecture

- 7.1 Regulation Loader & Auto-Classification

7.2 Continuous Monitoring & Incident Reporting

7.3 Cross-Jurisdiction Mapper

8. Integration Architecture

9. Infrastructure & Deployment

10. Technology Stack Summary

1. Executive Summary

Adhi Compliance is an AI-powered compliance management platform developed by **RootedAI** that enables organizations deploying artificial intelligence to navigate the rapidly evolving global regulatory landscape with confidence.

As AI regulation intensifies — the EU AI Act mandating fines up to **€35 million**, over **700 AI-related bills** across 45 US states, and emerging frameworks in India and China — organizations face an unprecedented compliance burden with virtually zero purpose-built infrastructure.

Adhi Compliance provides:

- **Automated Risk Classification** — auto-classify AI systems per EU AI Act risk tiers (Unacceptable, High, Limited, Minimal) and other jurisdictional frameworks
- **Continuous Compliance Monitoring** — real-time tracking across EU, US, India, and China regulatory regimes with proactive alerting
- **Bias & Fairness Auditing** — automated bias detection in AI model outputs using statistical and ML methods
- **Transparency Documentation** — automated model cards, data sheets, and impact assessments
- **Incident Reporting** — streamlined EU AI Act Article 62 workflows with deadline tracking
- **Cross-Jurisdiction Mapping** — unified compliance view across multiple regulatory regimes

> **Heritage:** Adhi Compliance evolved from a multimodal RAG system capable of ingesting video, audio, PDF, and image data for semantic search. This ingestion pipeline is retained as foundation for processing complex regulatory documents and audit artifacts.

2. System Architecture Overview

The platform follows a modern three-tier architecture designed for horizontal scalability and multi-tenant SaaS deployment.

Architectural Principles

- **API-First** — all functionality exposed via versioned REST endpoints; the frontend is a first-party consumer

- **Event-Driven** — long-running tasks handled asynchronously via Celery + Redis
- **Multi-Tenant Isolation** — Row-Level Security at database level for complete data isolation
- **AI-Native** — LLM and embedding models are core infrastructure, not bolt-ons
- **Regulation-as-Data** — frameworks are structured, versioned, queryable datasets

3. Frontend Architecture

3.1 Application Structure & Key Pages

Built on **Next.js 16** (App Router, React Server Components, TypeScript).

Page Breakdown

Page	Purpose	Key Components
Dashboard	Executive compliance overview	ComplianceScoreRing, RiskHeatmap, AlertsFeed, SystemStatusGrid

AI Systems	Catalog all deployed AI systems	SystemCard, RiskBadge, ClassificationWizard, SystemForm
Regulations	Browse & search regulations	RegulationBrowser, ArticleViewer, ApplicabilityChecker
Compliance Chat	RAG-powered regulatory Q&A	ChatInterface, SourceCitation, ContextPanel
Bias Auditor	Run & review fairness audits	AuditConfigForm, MetricsChart, BiasReport, DataUploader
Reports	Generate model cards & exports	ReportBuilder, PDFPreview, TemplateSelector
Incidents	EU AI Act Article 62 reporting	IncidentForm, TimelineView, StatusTracker
Settings	Org config, team, integrations	CompanyForm, InviteModal, WebhookConfig, APIKeyManager

3.2 Component Hierarchy & State Management

- **Server State** — TanStack Query v5: caching, background refetching, optimistic updates
- **Client State** — React Context (auth, theme, sidebar) + Zustand (complex UI state)
- **URL State** — Next.js searchParams for filters, pagination, shareable views
- **Real-time** — Supabase Realtime for live compliance alerts

3.3 Design System

Element	Specification
Primary Background	`rgb(5,5,5)` with dotted grid overlay
Surface Cards	`rgba(255,255,255,0.06)` + `backdrop-blur(12px)`

Primary Accent	`rgb(59,130,246)` blue → `rgb(168,85,247)` purple gradients
Typography	Outfit (headings), Montserrat (body), JetBrains Mono (code)
Border Radius	12px cards, 50px pills/buttons, 6px inputs
Animations	Framer Motion: fade-up, spring transitions, number counters
Icons	Lucide React
Charts	Recharts with custom dark theme

4. Backend Architecture

4.1 FastAPI Service Layer & API Routes

Complete API Endpoint Reference

Method	Endpoint	Description
POST	`/api/v1/auth/register`	Register user + company
POST	`/api/v1/auth/login`	Authenticate → JWT
POST	`/api/v1/auth/refresh`	Refresh access token
GET	`/api/v1/auth/me`	Current user profile
GET	`/api/v1/companies/{id}`	Company profile
PUT	`/api/v1/companies/{id}`	Update company

GET	`/api/v1/ai-systems`	List AI systems (paginated, filterable)
POST	`/api/v1/ai-systems`	Register new AI system
GET	`/api/v1/ai-systems/{id}`	System detail + compliance
PUT	`/api/v1/ai-systems/{id}`	Update system metadata
DEL	`/api/v1/ai-systems/{id}`	Decommission system
POST	`/api/v1/ai-systems/{id}/classify`	Trigger risk classification
GET	`/api/v1/regulations`	List regulations (filterable)
GET	`/api/v1/regulations/{id}`	Regulation detail + articles
POST	`/api/v1/regulations/search`	Semantic vector search (RAG)
POST	`/api/v1/compliance/check`	Run compliance check
GET	`/api/v1/compliance/status/{sys_id}`	Compliance status per system
GET	`/api/v1/compliance/dashboard`	Aggregated metrics
GET	`/api/v1/compliance/deadlines`	Upcoming regulatory deadlines
POST	`/api/v1/bias-audits`	Initiate bias audit (async)
GET	`/api/v1/bias-audits/{id}`	Audit results
GET	`/api/v1/bias-audits`	List audits
POST	`/api/v1/incidents`	Report incident
GET	`/api/v1/incidents`	List incidents
PUT	`/api/v1/incidents/{id}`	Update incident status
POST	`/api/v1/reports/generate`	Generate report (async)
GET	`/api/v1/reports/{id}/download`	Download report file
POST	`/api/v1/chat`	Compliance chat (streaming)
POST	`/api/v1/webhooks`	Configure webhook
GET	`/api/v1/webhooks`	List webhooks
DEL	`/api/v1/webhooks/{id}`	Remove webhook

4.2 Authentication & Authorization

- **Supabase Auth (GoTrue)** — registration, login, password reset, OAuth (Google/GitHub)
- **JWT Tokens** — short-lived access (1h) + refresh tokens, RS256 signed
- **Row-Level Security** — PostgreSQL RLS policies enforce company-level isolation at the DB level
- **RBAC Roles:** `owner` · `admin` · `analyst` · `viewer` — enforced via FastAPI dependencies

4.3 Background Task Processing

5. Data Architecture

5.1 Complete Database Schema

companies

Column	Type	Description
id	UUID PK	Primary key
name	VARCHAR(255)	Company name
industry	VARCHAR(100)	Industry sector
size	ENUM	startup / sme / enterprise
headquarters_country	VARCHAR(2)	ISO country code
operating_jurisdictions	JSONB	Array of jurisdiction codes
subscription_tier	ENUM	starter / growth / enterprise
created_at	TIMESTAMPTZ	Creation timestamp
updated_at	TIMESTAMPTZ	Last update

users

Column	Type	Description
id	UUID PK	Refs Supabase auth.users
company_id	UUID FK	→ companies.id
email	VARCHAR(255)	User email
full_name	VARCHAR(255)	Display name
role	ENUM	owner / admin / analyst / viewer
is_active	BOOLEAN	Account status
last_login	TIMESTAMPTZ	Last login

ai_systems

Column	Type	Description
id	UUID PK	Primary key
company_id	UUID FK	→ companies.id
name	VARCHAR(255)	System name
description	TEXT	Purpose & functionality
system_type	VARCHAR(100)	recommendation / classification / generative / etc.
deployment_status	ENUM	development / staging / production / retired
risk_level	ENUM	unacceptable / high / limited / minimal / unclassified
risk_classification_details	JSONB	Detailed reasoning from classifier
model_details	JSONB	Model type, version, provider, parameters
data_sources	JSONB	Training/input data descriptions
affected_populations	JSONB	Impacted groups
jurisdictions	JSONB	Deployment jurisdictions
owner_user_id	UUID FK	→ users.id (responsible person)
created_at / updated_at	TIMESTAMPTZ	Timestamps

regulations

Column	Type	Description
id	UUID PK	Primary key
name	VARCHAR(255)	e.g. "EU AI Act"
jurisdiction	VARCHAR(50)	EU / US_FEDERAL / US_STATE_CA / IN / CN
status	ENUM	draft / enacted / effective / superseded
effective_date	DATE	When regulation takes effect
summary	TEXT	Plain-language summary
full_text	TEXT	Complete text
key_requirements	JSONB	Structured requirements

penalties	JSONB	Penalty structures
source_url	VARCHAR(500)	Official source
version	INTEGER	Version number

regulation_chunks (for RAG)

Column	Type	Description
id	UUID PK	Primary key
regulation_id	UUID FK	→ regulations.id
chunk_text	TEXT	Text chunk (≤512 tokens)
article_ref	VARCHAR(50)	Article/section reference
embedding	VECTOR(384)	MiniLM-L6-v2 embedding
metadata	JSONB	Chunk metadata

compliance_checks

Column	Type	Description
id	UUID PK	Primary key
ai_system_id	UUID FK	→ ai_systems.id
regulation_id	UUID FK	→ regulations.id
check_type	ENUM	manual / automated / scheduled
status	ENUM	compliant / non_compliant / partial / pending
score	FLOAT	Compliance score 0–100
findings	JSONB	Detailed findings array
recommendations	JSONB	Remediation steps
checked_by	UUID FK	→ users.id
checked_at	TIMESTAMPTZ	When check ran
next_check_due	TIMESTAMPTZ	Next scheduled check

bias_audits

Column	Type	Description
id	UUID PK	Primary key
ai_system_id	UUID FK	→ ai_systems.id
audit_type	ENUM	demographic_parity / equalized_odds / disparate_impact / custom
status	ENUM	pending / running / completed / failed
dataset_info	JSONB	Dataset description, size, features
protected_attributes	JSONB	Attributes tested (gender, race, age...)
results	JSONB	Metrics, scores, visualization data
recommendations	JSONB	Mitigation recommendations
initiated_by	UUID FK	→ users.id
started_at / completed_at	TIMESTAMPTZ	Timing

incidents

Column	Type	Description
id	UUID PK	Primary key
ai_system_id	UUID FK	→ ai_systems.id
title	VARCHAR(255)	Incident title
description	TEXT	Detailed description
severity	ENUM	critical / high / medium / low
incident_type	ENUM	safety / bias / privacy / security / performance
status	ENUM	reported / investigating / resolved / closed
affected_parties	JSONB	Affected individuals/groups
root_cause	TEXT	Root cause analysis
remediation_actions	JSONB	Actions taken
reported_to_authority	BOOLEAN	Reported per Art. 62?
authority_report_date	TIMESTAMPTZ	Date reported
reported_by	UUID FK	→ users.id

occurred_at	TIMESTAMPTZ	When incident occurred

Additional Tables

Table	Key Fields	Purpose
compliance_deadlines	regulation_id, deadline_date, description, status	Track regulatory deadlines
model_cards	ai_system_id, content (JSONB), version, generated_at	Auto-generated model cards
audit_logs	user_id, action, entity_type, entity_id, before/after (JSONB), timestamp	Full audit trail
webhook_configs	company_id, url, events (JSONB), secret, is_active	Webhook configuration
notification_preferences	user_id, channel, event_types, is_enabled	Per-user notification settings

5.2 Vector Store & Data Flow

6. AI/ML Architecture

6.1 LLM Integration — Google Gemini

Gemini 1.5 Flash powers multiple subsystems:

Subsystem	LLM Use Case	Strategy
Risk Classification	Analyze system against EU AI Act Annex III	Structured JSON output + chain-of-thought
Compliance Chat	RAG-powered regulatory Q&A	System prompt + retrieved chunks + query
Bias Interpretation	Interpret statistical metrics → recommendations	Metrics summary → plain-language analysis
Model Card Gen	Generate cards from system metadata	Template filling + contextual expansion
Incident Analysis	Severity assessment & reporting guidance	Classification + obligation mapping
Regulation Summary	Summarize new regulations	Long-context extraction of key requirements

6.2 Risk Classification Engine

6.3 Bias Detection & Embedding Pipeline

- **Demographic Parity** — $P(\hat{Y}=1|A=0) = P(\hat{Y}=1|A=1)$; equal positive rates across groups
- **Equalized Odds** — equal TPR and FPR across groups
- **Disparate Impact Ratio** — selection rate ratio (<0.8 = potential disparate impact)
- **Calibration** — predicted probabilities match actuals per group

7. Compliance Engine Architecture

7.1 Regulation Loader & Auto-Classification

The regulation loader ingests from multiple sources:

- **Official Gazettes** — EU Official Journal, India Gazette, US Federal Register
- **Legislative Databases** — EUR-Lex, congress.gov, PRS India
- **Curated Packages** — manually verified regulation bundles

Pipeline: text extraction → structural parsing → requirement extraction → embedding → cross-reference linking.

Auto-classification on new AI system registration:

1. Keyword matching against prohibited use patterns (Art. 5)
2. Category matching against Annex III high-risk list
3. LLM analysis for ambiguous cases (structured JSON output)
4. Confidence scoring — low-confidence flagged for human review
5. Multi-jurisdiction mapping for all applicable regions

7.2 Continuous Monitoring & Incident Reporting

Incident Reporting Workflow (EU AI Act Article 62)

1. **Detection** — user report or automated anomaly trigger
2. **Severity Assessment** — LLM-assisted triage: "serious incident" per Art. 62?
3. **Data Collection** — guided form: system ID, description, affected parties, timeline
4. **Authority Mapping** — identifies relevant authority (AI Office for EU, MEITY for India)
5. **Report Generation** — structured report in required format
6. **Deadline Tracking** — 15-day submission deadline with automated reminders
7. **Follow-up** — manages required follow-up reports and resolution

7.3 Cross-Jurisdiction Mapper

Requirement	EU AI Act	US (NIST AI RMF)	India (DPDP)	China (AIGC)
Risk Assessment	Art. 9 — mandatory (high-risk)	MAP function — recommended	DPIA for significant data	Algorithm assessment required
Transparency	Art. 13 — mandatory disclosure	GOVERN 4 — recommended	Notice to data subjects	Content labeling required
Human Oversight	Art. 14 — mandatory (high-risk)	GOVERN 1 — recommended	Not explicit	Manual review for recs
Incident Reporting	Art. 62 — 15 days	Voluntary (sector-specific)	CERT-In: 6 hours	CAC reporting required
Bias/Fairness	Art. 10 — data governance	MEASURE 2 — recommended	Non-discrimination	Anti-discrimination req.

8. Integration Architecture

Webhook System

- **Events:** compliance_check.completed · risk_level.changed · deadline.approaching · incident.reported · regulation.updated · bias_audit.completed
- **Targets:** Slack (incoming webhook) · Microsoft Teams (connector) · Email (SendGrid) · Custom HTTP
- **Delivery:** at-least-once, exponential backoff retry (3 attempts)
- **Security:** HMAC-SHA256 signature verification

API-First Integrations

- **CI/CD** — compliance checks in ML model deployment pipelines
- **SIEM** — push compliance events to security platforms
- **GRC Platforms** — sync with enterprise governance tools
- **Custom Dashboards** — build reporting on top of our API

Export Engine

Format	Use Case	Technology
PDF	Model cards, compliance reports, audit reports	WeasyPrint / Puppeteer
Excel	Compliance matrices, audit data	openpyxl
JSON	API responses, data exchange	Native FastAPI
CSV	Bulk export, regulatory mappings	Python csv

9. Infrastructure & Deployment

Cloud Architecture

Docker Containerization

Security Architecture

- **Encryption at Rest** — AES-256 (Supabase/AWS managed)
- **Encryption in Transit** — TLS 1.3 everywhere
- **Row-Level Security** — PostgreSQL RLS for tenant isolation
- **API Auth** — JWT with RS256, short-lived tokens (1h)
- **Rate Limiting** — Redis sliding window per API key
- **Audit Logging** — all compliance actions logged with user, timestamp, before/after
- **Secrets** — AWS Secrets Manager (never in code)
- **OWASP Top 10** — input validation, CSRF, SQLi prevention (ORM), XSS (React)

10. Technology Stack Summary

Layer	Technology	Version	Purpose
Frontend	Next.js	16.x	React framework (App Router)
	TypeScript	5.x	Type safety
	Tailwind CSS	4.x	Utility-first styling
	Framer Motion	11.x	Animations
	TanStack Query	5.x	Server state management
	Recharts	2.x	Data visualization
	Lucide React	latest	Icons
Backend	Python	3.10+	Primary language
	FastAPI	0.109+	API framework
	SQLAlchemy	2.x	ORM
	Alembic	1.x	Migrations
	Pydantic	2.x	Validation & schemas
	Celery	5.x	Distributed task queue
AI/ML	Google Gemini	1.5 Flash	LLM compliance analysis
	sentence-transformers	MiniLM-L6-v2	Embeddings (384-dim)
	Sarvam AI	latest	Indian language transcription
Database	PostgreSQL	15+	Primary DB (Supabase)
	pgvector	0.5+	Vector search
	Redis	7.x	Cache + queue
Auth	Supabase Auth	GoTrue	Auth + JWT
Storage	Supabase Storage	S3-compat	Files & reports
Infra	Docker	24.x	Containerization

	AWS ECS Fargate	—	Orchestration
	Vercel	—	Frontend hosting
	CloudFront	—	CDN
Monitoring	Sentry	latest	Error tracking
	LogRocket	latest	Session replay
	CloudWatch	—	Infra monitoring
CI/CD	GitHub Actions	—	Build & deploy